WORKED SOLUTION

STEP III 2024, Question 7

Pure • Irrationality of e Squared

STEP III 2024, Question 7

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2024 STEP papers (a zip file):

www.ocr.org.uk/Images/725753-2024-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 1.4, *e Is Irrational, and So Is e^2*: https://beyondstep.keithmcnulty.org/ch01/e-irrational.html

The key idea

Both series are the tails of the exponential series once it has been multiplied by a factorial. Multiply $\mathrm{e} = \sum 1/r!$ by $(2n)!$ and every term up to $r = 2n$ becomes an integer, while everything after is exactly $\mathrm{f}(2n)$; the same happens for $1/\mathrm{e}$ and $\mathrm{g}(2n)$. If e^2 were rational, a combination of these small tails would be forced to be an integer, yet it is squeezed strictly between 0 and 1.

Solution

Write

$$a_k = \frac{1}{(n+1)(n+2)\cdots(n+k)}, \qquad k \geq 1,$$

so that $\mathrm{f}(n) = a_1 + a_2 + a_3 + \cdots$ and $\mathrm{g}(n) = a_1 - a_2 + a_3 - \cdots$. Note that $a_{k+1} = a_k/(n+k+1)$.

Part (i)

Every a_k is positive, so $\mathrm{f}(n) > 0$. Each factor in the denominator of a_k is at least $n+1$, so $a_k \leq 1/(n+1)^k$, with strict inequality once $k \geq 2$ (the factor $n+2$ appears). Comparing with a geometric series of ratio $\frac{1}{n+1} < 1$,

$$\mathrm{f}(n) < \sum_{k=1}^{\infty} \frac{1}{(n+1)^k} = \frac{\frac{1}{n+1}}{1 - \frac{1}{n+1}} = \frac{1}{n}.$$

Hence $0 < \mathrm{f}(n) < \frac{1}{n}$.

Part (ii)

Consecutive terms strictly decrease: $a_{k+1} = a_k/(n+k+1) < a_k$, since $n+k+1 > 1$. Grouping in pairs from the start,

$$\mathrm{g}(n) = (a_1 - a_2) + (a_3 - a_4) + \cdots > 0,$$

as each bracket is positive. Grouping instead after the first term,

$$\mathrm{g}(n) = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots < a_1 = \frac{1}{n+1},$$

as each bracket is again positive. So $0 < \mathrm{g}(n) < \frac{1}{n+1}$.

Part (iii)

Let $m = 2n$. Using $\mathrm{e} = \sum_{r\geq 0} 1/r!$,

$$m!\,\mathrm{e} = \sum_{r=0}^{m} \frac{m!}{r!} + \sum_{r=m+1}^{\infty} \frac{m!}{r!}.$$

For $r \leq m$, $m!/r! = (r+1)(r+2)\cdots m$ is an integer, so the first sum is an integer; call it A. In the second sum, $m!/(m+k)! = 1/\big((m+1)\cdots(m+k)\big)$, which is the kth term of $\mathrm{f}(m)$. So $m!\,\mathrm{e} = A + \mathrm{f}(m)$, and $(2n)!\,\mathrm{e} - \mathrm{f}(2n) = A$ is an integer.

Similarly $1/\mathrm{e} = \sum_{r\geq 0}(-1)^r/r!$ gives

$$\frac{m!}{\mathrm{e}} = \underbrace{\sum_{r=0}^{m}(-1)^r \frac{m!}{r!}}_{B\in\mathbb{Z}} + \sum_{k=1}^{\infty} \frac{(-1)^{m+k}}{(m+1)(m+2)\cdots(m+k)}.$$

? **Ask yourself** Why does the question use $2n$ rather than n?

Because $m = 2n$ is even, $(-1)^{m+k} = (-1)^k$, so the tail is $-a_1 + a_2 - a_3 + \cdots = -\mathrm{g}(m)$ (with a_k now formed from m). Thus $m!/\mathrm{e} = B - \mathrm{g}(m)$, and $\dfrac{(2n)!}{\mathrm{e}} + \mathrm{g}(2n) = B$ is an integer. (For odd m the tail would be $+\mathrm{g}(m)$ and we would need a minus sign instead.)

Part (iv)

Using the integers A and B from part (iii),

$$\begin{aligned} q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n) &= q\big((2n)!\,\mathrm{e} - A\big) + p\Big(B - \frac{(2n)!}{\mathrm{e}}\Big) \\ &= (2n)!\Big(q\mathrm{e} - \frac{p}{\mathrm{e}}\Big) - qA + pB \\ &= pB - qA, \end{aligned}$$

since $q\mathrm{e} = p/\mathrm{e}$. As p, q, A, B are integers, so is $q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n)$, for every positive integer n.

Part (v)

Suppose, for contradiction, that e^2 is rational. Since $\mathrm{e}^2 > 0$ we can write $\mathrm{e}^2 = p/q$ with p, q positive integers, and dividing by e gives $q\mathrm{e} = p/\mathrm{e}$. By part (iv), $N_n = q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n)$ is an integer for every $n \geq 1$.

? **Ask yourself** How can we make N_n fail to be an integer?

Squeeze it with parts (i) and (ii), applied at $2n$. Since $p, q > 0$,

$$0 < N_n < \frac{q}{2n} + \frac{p}{2n+1} < \frac{p+q}{2n}.$$

Now choose n large enough that $2n > p + q$. Then $0 < N_n < 1$, so N_n is an integer strictly between 0 and 1, which is impossible. Hence e^2 is irrational.

Checks

Small cases can be summed exactly. For $n = 2$, part (iii) says $24\mathrm{e} - \mathrm{f}(4)$ is an integer; indeed $\sum_{r=0}^{4} 24/r! = 65$, and $\mathrm{f}(4) = 24\mathrm{e} - 65 \approx 0.2388$, comfortably inside $(0, \frac{1}{4})$. Likewise $\mathrm{f}(2) = 2\mathrm{e} - 5 \approx 0.4366 < \frac{1}{2}$ and $\mathrm{g}(2) = 1 - 2/\mathrm{e} \approx 0.2642 < \frac{1}{3}$. A computer check confirmed the bounds of (i) and (ii) for $n \leq 24$, the integrality in (iii) for $2n \leq 22$, and that $0 < q\,\mathrm{f}(2n) + p\,\mathrm{g}(2n) < (p+q)/(2n)$ for random p, q.

Common slips

- Claiming $a_k < 1/(n+1)^k$ for all k: equality holds at $k = 1$, so the strict inequality comes from $k \geq 2$.
- Forgetting the sign of the tail in the $1/\mathrm{e}$ expansion: it is $-\mathrm{g}(2n)$ only because $2n$ is even.
- Stopping at "N_n is small": the contradiction needs $N_n > 0$ as well, which is why the lower bounds in (i) and (ii) matter.

Beyond STEP

This is Fourier's classic proof that e is irrational, extended by splitting $\mathrm{e}^2 = p/q$ into $q\mathrm{e} = p/\mathrm{e}$ so that both series converge fast. The same idea, with more care, shows e^r is irrational for every non-zero rational r, and Liouville used it to show that e is not a root of any quadratic with integer coefficients. Bounding tails of series is a core technique of first-year analysis; see Abbott, *Understanding Analysis*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.